

1.1 System Requirement Specifications (SRS)

1. Introduction

- Purpose:

This document is a document that outlines the system requirements of SkillXChange, which is a web-based skills barter system where individuals can acquire or give skills without the use of money. The primary idea behind the system is to encourage peer-to-peer-learning, collaboration and knowledge sharing through the provision of safety and orderly environment of skills. This specification is aimed at the developers, testers, supervisors, and the academic reviewers to learn the functionality and limitations of the system.

- Scope
 - User registration and login system.
 - User profile and skill management (skills to teach and learn).
 - Browsing and matching skills between users.
 - Request and approval of skill exchange.
 - User-to-user communication during skill exchange.
 - Post-skilling feedback and rating.
- Definitions, Acronyms, and Abbreviations
 - SRS – Software Requirements Specification
 - UML – Unified Modeling Language
 - DFD – Data Flow Diagram
 - ERD – Entity Relationship Diagram

2. Overall Description

- Product Perspective:

Skill Barter System is an independent web-based application that is meant to facilitate skill exchange without any monetary transactions. The system is used by the users with the Web interface, and all the information concerning the users, skills, exchanges, and feedback is stored and managed in the central database. The system is client-server based on the MVC pattern of design.

- User Classes and Characteristics
 - Admin
 - Manage user
 - Manage skills
 - Monitor activity
 - Handle policy
 - User
 - Register and Log In
 - Manage profile
 - Adds skills to teach and learn
 - Skill exchange
 - Provide feedback
- Operating Environment
 - Web browsers
 - Internet connection
 - Server with PHP (Laravel Framework)
 - MySQL database
- Design and Implementation Constraints
 - System must be web-based
 - Internet connection required
 - Data must store
 - Only registered users can access system features

3. Functional Requirements

- User Authentication
 - The system should enable its users to register using valid information.
 - The system will enable users to log in with email/username and password.
 - The system must ensure that the credentials provided by the user are validated before access is allowed.
- Profile and Skill Management

- The system should enable users to create and modify their profiles.
- The system should enable users to enter, modify, and remove skills.
- The system should be capable of storing skill information like category and description.
- Skill Browsing and Matching
 - The system should enable browsing skills by users.
 - The system should support searching for skills based on categories or keywords.
 - Users will see relevant skill correlations within the system.
- Skill Exchange Requests
 - The system will enable the sending of skill exchange requests.
 - It shall be possible in this system for a user to accept or reject requests.
 - The system should record the status of every exchange.
- Communication
 - The system should enable users to communicate within the system.
 - Messages will be associated with exchange sessions.
- Feedback and Rating
 - The system shall enable feedback from users following an exchange.
 - The system shall be capable of storing ratings based on user reliability.
 - Admin should be able to view feedback.

4. Non-functional Requirements

- Performance
 - The system will take 2-3 seconds to load pages.
 - The system must support several users at once.
- Security
 - Password must be encrypted for users
 - Role-Based Access Control must be implemented by the system.
 - Secure protocols (HTTPS) must be used for information transport.
- Usability
 - The system should be equipped with an easy-to-use interface
 - Both desktop computers and mobile phones should be able to access the system.
 - The error messages will be simple to read and comprehend.
- Reliability
 - The data integrity will be supported by the system.
 - Minor errors will be properly corrected by the system.

5. Future Enhancements

- AI-based skill matching
- Gamification and rewards system
- Premium membership
- Advanced analytics & reporting